

NS-2026-0119: Mapping the Evidence Gap Between NMN and NR for Metabolic Outcomes: A Systematic Review, Transitivity Assessment, and Indirect Comparison Meta-Analysis

Reviewer Comments to the Authors

Recommendation: Major Revision

This manuscript presents a systematic review and indirect comparison meta analysis evaluating nicotinamide mononucleotide (NMN) and nicotinamide riboside (NR) for metabolic outcomes. The topic is timely, relevant, and of considerable interest to nutritional science, given the rapid growth of NAD⁺ precursor supplementation and the absence of direct comparative trials. The manuscript is methodologically ambitious and demonstrates commendable transparency through PROSPERO registration, PRISMA adherence, dual review procedures, RoB2 assessment, GRADE/CINeMA evaluation, and open source availability of the analytical code. The transitivity focused framing is novel and potentially valuable to the field.

However, several major methodological and interpretive issues substantially limit confidence in the indirect comparative findings in the current form. I believe the manuscript has potential for publication after substantial revision.

Major Comments

1. Fundamental concerns regarding transitivity and validity of indirect comparison

A major concern relates to the assumption of transitivity/exchangeability, which underpins the validity of the Bucher indirect comparison and appears substantially challenged in the current evidence base. The manuscript appropriately identifies considerable asymmetry between the NMN and NR literature, including major molar dose differences (approximately 1.9–9.2-fold), systematic geographic and ethnic variation between study populations, differences in baseline metabolic risk, inclusion of disease specific NR populations (e.g., chronic kidney disease, COPD, Parkinson's disease, and mild cognitive impairment) without equivalent NMN comparators, and incompatible NAD⁺ biomarker assays. These issues are not minor methodological differences but represent important threats to comparability and the validity of indirect inference. In its current form, the manuscript presents a degree of conceptual tension in which the evidence base is described as structurally unsuitable for reliable indirect comparison, yet pooled indirect estimates are quantitatively interpreted. The authors should address this issue more explicitly and consider reframing the manuscript away from an emphasis on comparative efficacy between NMN and NR toward mapping the evidence gap with exploratory indirect comparison. Correspondingly, the interpretation of indirect estimates should be substantially more cautious and framed within the considerable uncertainty and limitations of the available evidence.

2. Overinterpretation of null findings

The manuscript would benefit from more cautious interpretation of null findings, as the current wording occasionally implies equivalence or lack of effect between NMN and NR. Statements such as Neither precursor significantly improved most outcomes, and no indirect comparison reached statistical significance risk conflating the absence of evidence with evidence of absence. Given the substantial limitations of the evidence base, including very small sample sizes, short intervention durations, considerable heterogeneity across populations and study designs, low statistical power, and uniformly Very Low certainty evidence, the findings should be interpreted more conservatively. The conclusions would be strengthened by using language that better reflects the underlying uncertainty, for example, "Current evidence is insufficient to establish meaningful differences between NMN and NR," rather than wording that may imply no meaningful difference exists. Such revision would provide a more balanced and scientifically appropriate interpretation of the available evidence.

3. Statistical methodology requires stronger justification

The statistical methodology would benefit from stronger justification and greater transparency. The manuscript relies primarily on the DerSimonian–Laird (DL) random effects estimator; however, given the sparse evidence base, with many outcomes informed by only 1–4 studies, the appropriateness of this approach warrants further consideration. In small meta-analyses, DL estimation may be unstable and may underestimate uncertainty, potentially leading to overconfident confidence intervals. Although the authors indicate that the results were validated using restricted maximum likelihood (REML) in R, this validation is not adequately described in the Results section, and its impact on the findings remains unclear. The authors should therefore clarify the rationale for selecting DL as the primary estimator, indicate whether Hartung–Knapp adjustments were considered to better account for uncertainty in small meta analyses, and explicitly state whether REML materially altered conclusions. In addition, reporting sensitivity analyses using alternative estimators would strengthen confidence in the robustness of the findings, particularly given the limited number of included studies and the already substantial uncertainty surrounding the evidence base.

4. Excessive reliance on single-study outcomes

The manuscript appropriately acknowledges that 17 of 36 pairwise comparisons were based on a single study; however, this limitation warrants greater emphasis throughout the manuscript. In such cases, the use of the term meta analysis may be misleading, as these estimates reflect individual trial findings rather than a pooled quantitative synthesis. The authors should provide a clearer distinction between outcomes derived from true pooled meta-analytic estimates and those based on single study evidence, as the level of inferential confidence differs substantially between the two. Stronger caveats regarding interpretability are also needed, particularly where conclusions may appear more robust than the underlying evidence supports. In addition, consideration

should be given to moving some sparsely informed outcomes, especially those based solely on single studies, to supplementary material to improve focus and avoid overstating the strength of evidence in the main manuscript.

5. Clinical heterogeneity may be too extensive for pooled interpretation

The included NR trials encompass highly heterogeneous populations, including individuals with Parkinson's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), mild cognitive impairment, hypertension, and healthy ageing adults, whereas the NMN studies predominantly involve healthy, overweight, or metabolically at-risk adults. This substantial clinical heterogeneity raises concerns about the appropriateness of pooling interpretations across markedly different metabolic and disease contexts and may further weaken the transitivity assumption underpinning the indirect comparison. Disease-specific populations may have distinct metabolic pathways, inflammatory states, medication use, and baseline NAD⁺ dynamics that could influence responsiveness to supplementation and limit comparability with generally healthy NMN cohorts. The authors should therefore provide stronger justification for retaining disease specific populations in the quantitative synthesis, clarify whether subgroup analyses by metabolic health status were considered or feasible, and discuss whether restricting analyses to more metabolically comparable populations would materially alter the findings. Addressing these issues would strengthen confidence in the interpretability and validity of the comparative conclusions.

6. Clarify handling of crossover studies

Several of the included NR studies employed crossover designs; however, the methodological handling of these studies requires clearer explanation in the Methods section. Specifically, the authors should clarify the assumptions used regarding within-subject correlation and how these were incorporated into effect size estimation and variance calculations. Greater detail is also needed regarding the assessment and management of potential carryover effects, particularly given the metabolic nature of the outcomes and the possibility of residual biological effects following supplementation. In addition, the adequacy of washout periods should be discussed and justified, especially where short washout durations may have been insufficient to eliminate lingering intervention effects. Finally, it would be helpful to clarify whether crossover studies were analysed separately, incorporated alongside parallel trials using adjusted methods, or explored through sensitivity analyses. Improved reporting of these methodological considerations would enhance transparency and confidence in the robustness of the findings.

7. Multiplicity adjustment needs stronger rationale

The manuscript reports the use of both Bonferroni and Benjamini–Hochberg multiplicity corrections across 14 indirectly comparable outcomes; however, the rationale for applying these adjustments requires further clarification. Specifically, the

authors should indicate whether multiplicity adjustment was prespecified in the study protocol or analytical plan, as post hoc implementation may influence interpretation of statistical findings. In addition, it would be helpful to clearly report whether any results changed materially following adjustment, including whether nominally significant findings lost significance or interpretation was otherwise altered. Finally, the appropriateness of multiplicity correction in the context of what appears to be an exploratory, hypothesis-generating analysis should be more explicitly justified, particularly given the small number of studies and already substantial limitations in certainty of evidence. Clarification of these issues would improve statistical transparency and aid interpretation of the reported findings.

8. Discussion should better contextualise biological plausibility

The discussion appropriately acknowledges several important methodological limitations; however, the biological plausibility and mechanistic interpretation of the findings require further development. Specifically, the manuscript would benefit from a more detailed discussion of the mechanistic differences between NMN and NR metabolism, including their distinct positions within the NAD⁺ salvage pathway and implications for bioavailability and metabolic effects. Greater consideration should also be given to plausible dose–response relationships, particularly whether the substantially higher molar dosing of NR across included trials may have biased indirect comparisons and affected interpretation of comparative efficacy. In addition, the discussion should more critically address the uncertainty surrounding the relevance of Slc12a8-mediated NMN transport in humans, as much of the mechanistic rationale derives from murine evidence and remains debated. Expanding discussion of these biological considerations would strengthen the nutritional and clinical interpretation of the findings and provide a more balanced contextualisation of the observed results.

9. Presentation and structure

The manuscript is generally well written but overly detailed in places.

I recommend:

Introduction: Reduce repetition on NAD⁺ biology and the rationale for supplementation.

Results: Avoid restating extensive numerical findings already shown in forest plots and supplementary tables.

Discussion: Streamline repeated emphasis on transitivity limitations.

The paper would benefit from improved conciseness.

Minor Comments

1. Please standardize terminology:

- Use either HbA1c or glycated haemoglobin consistently.
2. Clarify whether:
 - endpoint values,
 - change from baseline values,
 - or mixed extraction methods were used across studies.
 3. Table 2 (transitivity assessment) would benefit from:
 - clearer visual presentation,
 - explicit effect modifier categorisation.
 4. The rationale for selecting 2018 onward should be better justified.
 5. Consider including a brief paragraph on:
 - regulatory uncertainty surrounding NMN availability across jurisdictions.
 6. Please carefully proofread for minor stylistic repetition and overstatement.

Overall Recommendation

This manuscript addresses an important and clinically relevant question with notable methodological transparency and originality. However, major concerns regarding transitivity, indirect comparison validity, statistical robustness, and interpretation substantially limit confidence in the conclusions of the current version.

I encourage the authors to substantially revise the manuscript, particularly by reframing the work as an evidence gap and feasibility assessment, with a more cautious interpretation of comparative claims.

Recommendation: Major Revision